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BACHELOR'S THESIS
(Research Project)
Research Methods, Pre-training of Models for Text Analysis in Chemistry, Biochemistry and Related Fields

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Аннотация

В работе исследуется доменная адаптация предварительно обученных языковых моделей для задачи распознавания химических именованных сущностей (NER). Три кодировщика на основе архитектуры Transformer — BioBERT, многоязычный BERT (mBERT) и DeBERTa-v3 — сравниваются на корпусах BC5CDR и BC4CHEMD при едином рецепте дообучения. Далее применяется процедура доменно-адаптивного предобучения (DAPT) на химическом подъязыке и сравниваются пять стратегий дообучения (полное, частичная разморозка, обучение только токенизатора, обучение только классификационной головы, выборочная разморозка) с точки зрения качества и вычислительных затрат. Наилучшая конфигурация (BioBERT с DAPT на BC5CDR BC4CHEMD и последующим полным дообучением) достигает $F1 = 0{,}9098$ на BC5CDR, а DAPT снижает MLM-перплексию mBERT на химическом корпусе с 1,10×10 до 3,78. Дополнительно рецепт переносится на русский язык: на корпусе RuDReC дообучение mBERT достигает $F1 = 0{,}927$, что более чем втрое превышает zero-shot baseline ($F1 = 0{,}290$). Результаты показывают, что внутридоменный словарь и продолженное MLM-предобучение являются доминирующими факторами качества химического NER, тогда как конкретная стратегия дообучения оказывает вторичный эффект при фиксированном вычислительном бюджете; при наличии разметки на целевом языке вклад DAPT возрастает.

Abstract

This thesis investigates domain adaptation of pre-trained language models for the task of chemical Named Entity Recognition (NER). Three Transformer-based encoders — BioBERT, multilingual BERT (mBERT), and DeBERTa-v3 — are compared on the BC5CDR and BC4CHEMD corpora under a unified fine-tuning recipe. A domain-adaptive pre-training (DAPT) procedure is then applied on a chemical sub-language, and five fine-tuning strategies (full, partial unfreezing, tokenizer-only, classification-head-only, and selective unfreezing) are compared in terms of quality and computational cost. The best configuration (BioBERT with DAPT on BC5CDR BC4CHEMD followed by full fine-tuning) reaches F1 = 0.9098 on BC5CDR, while DAPT reduces the MLM perplexity of mBERT on a chemical corpus from 1.10×10 to 3.78. In addition, the recipe is transferred to Russian: on the RuDReC corpus, fine-tuning mBERT reaches F1 = 0.927, which is more than three times higher than the zero-shot baseline (F1 = 0.290). The results show that an in-domain vocabulary and continued MLM pre-training are the dominant factors for chemical NER performance, whereas the specific fine-tuning strategy has only a secondary effect under a fixed computational budget; when labeled data in the target language are available, the contribution of DAPT becomes more pronounced.

Contents

Key definitions, notation and abbreviations

The following acronyms and symbols are used throughout the thesis. They are defined here once and used without re-definition in later sections.

NER — Named Entity Recognition; the sequence-labelling task of detecting and classifying entity spans in text.

BIO — Begin–Inside–Outside tagging scheme used in this work with three tags: O, B-Chemical, I-Chemical.

MLM — Masked Language Modeling; the self-supervised objective used to pre-train BERT-style encoders.

DAPT — Domain-Adaptive Pre-Training; continued MLM training of a pre-trained encoder on an in-domain corpus before fine-tuning.

LM — Language Model.

PPL — Perplexity; geometric-mean reciprocal probability used here as the MLM evaluation metric.

BERT — Bidirectional Encoder Representations from Transformers.

mBERT — Multilingual BERT; the 104-language `bert-base-multilingual-cased` checkpoint.

BioBERT — Biomedical BERT; the `biobert-base-cased-v1.1` checkpoint, BERT-base continued on PubMed abstracts and PMC articles.

DeBERTa(-v3) — Decoding-enhanced BERT with disentangled attention (version 3 of the original architecture).

WordPiece / BPE / SentencePiece — sub-word tokenisation algorithms; mBERT and BioBERT use WordPiece, DeBERTa-v3 uses SentencePiece.

BC5CDR — BioCreative V Chemical-Disease Relation corpus; the primary English chemical-NER benchmark in Stages 1–2.

BC4CHEMD — BioCreative IV CHEMDNER corpus; the second English chemical corpus used in DAPT.

RuDReC — Russian Drug Reactions Corpus; the Russian-language drug-mention corpus used in Stage 3.

ADR — Adverse Drug Reaction (a RuDReC label, not used in our chemical-only label scheme).

DI — Drug Indication (a RuDReC label).

Drugname / Drugclass / Drugform — RuDReC drug-related labels; collapsed into a single Chemical class in Stage 3 to match the English experiments.

P, R, F1 — entity-level precision, recall and F1, computed in the strict `seqeval` mode (exact span and label match).

L0–L4 — the five ablation levers of Stage 3 (see Section 6.1).

GPU — Graphics Processing Unit; all experiments were run on a single Google Colab T4 GPU.

`[CLS]`, `[SEP]`, `[MASK]`, `[PAD]` — BERT special tokens.

$X, Y, \mathscr{Y}$ — input token sequence, output tag sequence and finite tag set, as defined in Section 2.1.

θ — trainable model parameters; the subset that is actually updated depends on the fine-tuning strategy (see Section 4.5).

1 Introduction

The past decade has been marked by a dramatic growth of scientific publications in chemistry, biochemistry and related disciplines. According to PubMed statistics, more than one million new biomedical articles are indexed every year, and a comparable flow of chemical literature is reported by CAS and ChemRxiv. Manual curation of such a stream is no longer realistic, and this fact motivates the development of automatic methods for extracting structured information from unstructured scientific text. Among these methods, Named Entity Recognition (NER) occupies a central place, because the identification of chemical compounds, drugs, diseases and genes in text is a prerequisite for downstream applications such as knowledge graph construction, relation extraction, drug–disease association mining and adverse-event surveillance.

Modern NER pipelines rely almost exclusively on pre-trained language models based on the Transformer architecture [1]. The original BERT model [2] and its multilingual extension (mBERT) demonstrated that bidirectional self-attention trained with a Masked Language Modeling (MLM) objective on a large unlabelled corpus can produce representations that transfer well to many downstream tasks after fine-tuning. However, the chemical and biomedical sublanguages differ substantially from the general-domain text used to pre-train mBERT: they are saturated with Greek letters, locants, stereodescriptors, IUPAC names, gene symbols and dosage expressions. Domain-specific models such as BioBERT [3], SciBERT [4] and PubMedBERT [5] were therefore proposed, and they consistently outperform the general-domain baselines on biomedical benchmarks. A complementary direction, known as Domain-Adaptive Pre-Training (DAPT) [6], keeps the model architecture unchanged but continues MLM pre-training on a targeted in-domain corpus before fine-tuning. DAPT has been reported as a lightweight, effective alternative to training a domain-specific model from scratch.

Despite the growing body of work on English biomedical NER, multilingual models such as mBERT have received comparatively little attention in this domain, even though a considerable share of chemical literature is published in languages other than English (Chinese, Russian, German, Japanese). This raises a practical question: how much can a general-purpose multilingual encoder be pushed towards chemical NER by domain-adaptive pre-training alone, and how does the result compare with dedicated biomedical models?

The present thesis contributes to this question along two lines. First, a controlled comparison of three encoders — BioBERT, mBERT and DeBERTa-v3 [7] — is carried out on chemical NER benchmarks under four fine-tuning strategies: frozen encoder, tokenizer-only training, partial unfreezing and full fine-tuning. The effect of continued MLM pre-training of mBERT on a chemical corpus is studied explicitly. Second, informed by the findings of the first stage, a deeper set of experiments is performed on BioBERT with the BC5CDR [8] and BC4CHEMD [9] corpora, including DAPT with different schedules and selective unfreezing. The resulting models are evaluated using precision, recall, F1 and MLM perplexity.

Goal of the thesis. The goal of this work is to evaluate, under a single controlled experimental setup, how much of the gap between a general multilingual encoder (mBERT) and a domain-specialised one (BioBERT) on biomedical chemical NER can be closed by domain-adaptive pre-training alone, and whether the same recipe carries over to a Cyrillic-script target language (Russian).

Tasks. To reach this goal, the following tasks are carried out:

1. formalise the chemical-NER task and the DAPT objective, and fix the evaluation protocol (Section 2);
2. review the literature on Transformer encoders, domain adaptation and biomedical NER, and position the thesis with respect to prior work (Section 3);
3. design and implement a uniform pipeline that supports five fine-tuning strategies on top of three encoders (Section 4);
4. run Stage 1–2 experiments on BC5CDR and BC4CHEMD and quantify the contribution of DAPT and of the fine-tuning strategy under a matched compute budget (Section 5);
5. extend the best Stage 2 recipe to Russian on RuDReC, isolate the contribution of supervised data versus cross-lingual DAPT, and report the resulting precision, recall and F1 (Section 6);
6. summarise the findings, state the practical takeaway and outline the natural follow-ups (Section 7).

Object and subject of the study. The *object* of the study is the family of pre-trained Transformer encoders applied to biomedical chemical NER. The *subject* is the effect of continued in-domain MLM pre-training (DAPT) and of different fine-tuning strategies on the quality of chemical-NER on English (BC5CDR, BC4CHEMD) and Russian (RuDReC) data.

Scientific novelty. The novelty of the work is twofold: (i) a controlled, same-budget comparison of mBERT, BioBERT and DeBERTa-v3 under five fine-tuning strategies and one DAPT recipe on the same chemical corpora; and (ii) a five-lever ablation that, for the first time on RuDReC, isolates the contribution of supervised in-language data from the contribution of cross-lingual English-only DAPT and from the contribution of raw in-language MLM data.

Reproducibility and code. All source code, training scripts and Colab notebooks used to produce the numbers reported in this thesis are available in the public repository `https://github.com/n-kozlovcev/VKR_DSBA`.

Structure of the thesis. The remainder of the thesis is organised as follows. Section 2 formalises the task and states the research questions. Section 3 reviews the literature on Transformer-based language models, domain adaptation and biomedical NER. Section 4 describes the experimental methods, architectures and training protocols. Section 5 presents and discusses the results of the English experiments (Stages 1–2). Section 6 extends the recipe to Russian on RuDReC (Stage 3). Section 7 concludes and outlines future work.

2 Problem Statement

2.1 Task formulation

Let $X = (x_1, x_2, \ldots, x_n)$ be a tokenised input sentence and let $\mathscr{Y}$ be a finite set of tag labels. The NER task consists in learning a function $f_\theta : \mathscr{V}^* \to \mathscr{Y}^*$ that maps X to a tag sequence $Y = (y_1, y_2, \ldots, y_n)$, $y_i \in \mathscr{Y}$, such that contiguous spans of tags encode named entities of interest. In this work the BIO tagging scheme is used, and the label set for chemical NER is

$$\mathscr{Y} = \{\text{O},\ \text{B-Chemical},\ \text{I-Chemical}\}, \tag{1}$$

where B-Chemical marks the first token of a chemical entity, I-Chemical marks its continuation, and O marks non-entity tokens.

The learning objective is the token-level categorical cross-entropy

$$\mathscr{L}_{\text{NER}}(\theta) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{c \in \mathscr{Y}} \mathbb{1}[y_i = c] \log p_\theta(y_i = c \mid X), \tag{2}$$

where N is the total number of sub-word tokens (excluding those masked as special or padding), $\mathbb{1}[\cdot]$ is the indicator function and $p_\theta(\cdot \mid X)$ is the probability distribution produced by the classification head on top of the encoder.

2.2 Domain-adaptive pre-training objective

For a subset of the experiments the encoder is additionally pre-trained on an unlabelled in-domain corpus $\mathscr{D}$ using the MLM loss

$$\mathscr{L}_{\text{MLM}}(\theta) = -\mathbb{E}_{X \sim \mathscr{D}} \sum_{i \in \mathscr{M}(X)} \log p_\theta(x_i \mid X_{\setminus \mathscr{M}}), \tag{3}$$

where $\mathscr{M}(X)$ is the set of positions masked in X, and $X_{\setminus \mathscr{M}}$ denotes the sequence with masked tokens replaced by the special symbol `[MASK]`. The quality of the adapted language model is summarised by the perplexity

$$\text{PPL}_{\text{MLM}} = \exp\bigl(\mathscr{L}_{\text{MLM}}\bigr). \tag{4}$$

2.3 Evaluation metrics

Per-class precision, recall and F1 are computed at the entity level and then micro-averaged across the two positive classes. For a given class c with true positives TP_c, false positives FP_c and false negatives FN_c,

$$P_c = \frac{TP_c}{TP_c + FP_c}, \quad R_c = \frac{TP_c}{TP_c + FN_c}, \quad F1_c = \frac{2P_cR_c}{P_c + R_c}. \tag{5}$$

The micro-averaged F1 is reported as the main quality criterion, complemented by loss and, for the DAPT experiments, by MLM perplexity.

2.4 Research questions and scope

The thesis addresses the following questions:

- Q1. How do a biomedical encoder (BioBERT), a multilingual encoder (mBERT) and a general-domain encoder with disentangled attention (DeBERTa-v3) compare on chemical NER under identical fine-tuning protocols?
- Q2. Does domain-adaptive pre-training of mBERT on a chemical corpus reduce the gap to dedicated biomedical models?
- Q3. Among frozen-encoder, tokenizer-only, partial-unfreezing and full fine-tuning, which strategy is most effective, and how sensitive is the outcome to the choice of encoder?
- Q4. Within a dedicated biomedical encoder (BioBERT), what is the marginal benefit of DAPT and selective unfreezing on top of full fine-tuning, and how does the benefit scale with corpus size and the number of MLM epochs?

The scope is limited to chemical and chemical–disease NER in English scientific text. Gene/protein NER, relation extraction and event extraction are outside the scope. The datasets used are BC5CDR, BC4CHEMD and ChemProt [10]. All experiments are run on a single NVIDIA GPU (Tesla T4 / A100 40 GB, depending on availability) using the Hugging Face Transformers library [11] and the `seqeval` evaluator [12].

3 Literature Review

3.1 From rule-based to neural NER

The history of Named Entity Recognition can be traced back to the Message Understanding Conferences (MUC) in the 1990s, where systems relied on hand-crafted rules, lexicons and gazetteers. Rule-based systems remained the dominant paradigm for more than a decade and are still used in production pipelines for tasks where high precision is more important than coverage, such as pharmacovigilance screening or regulatory compliance. Their main weakness is brittleness: small changes in writing style, punctuation or abbreviation conventions often require a significant engineering effort to accommodate.

The next generation of NER systems was based on shallow machine-learning models such as Hidden Markov Models, Maximum Entropy Markov Models and, most notably, Conditional Random Fields (CRF) [19]. A linear-chain CRF models the joint distribution of tag sequences conditioned on the input and captures label dependencies (for example the constraint that an `I-Chemical` tag cannot directly follow an `O` tag) through pairwise transition potentials. Feature engineering — part-of-speech tags, orthographic patterns, word shapes, lexicon look-ups and context windows — was the main source of improvements, and chemistry-specific features (presence of Greek letters, locants, stoichiometric numbers) were introduced by several groups working on early biomedical shared tasks.

The shift to deep learning started around 2015 with the bidirectional LSTM–CRF architecture of Lample et al. [20] and Huang et al. [21], which combined word-level embeddings (initialised from word2vec or GloVe), character-level convolutions and an LSTM context encoder. This combination reduced the need for hand-crafted features and became the de-facto baseline for NER until the arrival of Transformer-based encoders. The present thesis inherits from this line the idea of using sub-word representations as a bridge between form and meaning, but replaces the recurrent encoder with a pre-trained Transformer.

3.2 Word, sub-word and contextual representations

The quality of any sequence-labelling model is bounded by the quality of its input representation. Static word embeddings such as word2vec and GloVe provided the first scalable way to encode distributional similarity, but they assign a single vector to each word, which is problematic in chemistry where the same token (e.g. *lead*) can denote a metal, a verb or a compound name depending on context. Contextual embeddings from ELMo, produced by a bidirectional LSTM trained as a language model, were the first widely-used solution to polysemy and substantially improved biomedical NER benchmarks in 2018.

A parallel development was the introduction of sub-word tokenisation. Byte Pair Encoding (BPE) [22], WordPiece [23] and SentencePiece partition a word into shorter units based on corpus statistics, which is particularly useful for chemical text with its long compound names, hyphenated locants and Greek-letter prefixes. For example, the trisyllabic name *4-aminoquinoline* is split by the BioBERT tokenizer into `4`, `-`, `amino`, `##quin` and `##oline`, all of which occur frequently in PubMed and carry useful semantic information. Sub-word tokenisation also keeps the vocabulary size bounded (typically between 30k and 250k pieces) while still covering a very large number of surface forms, and it plays a central role in the mBERT / BioBERT comparison of this thesis: mBERT's 110k multilingual vocabulary has to encode chemical terms that belong to the English long tail, whereas BioBERT inherits the English BERT-cased vocabulary, which has been exposed to PubMed during continued pre-training.

3.3 Transformer-based language models

The Transformer architecture [1] replaced recurrent and convolutional encoders with multi-head self-attention, enabling efficient parallel computation and long-range dependency modelling. For an input representation $X \in \mathbb{R}^{n \times d}$, a scaled dot-product attention head computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{6}$$

where $Q = XW^Q$, $K = XW^K$, $V = XW^V$ are learned projections and d_k is the key dimension. Multiple heads compute independent projections in parallel, and their outputs are concatenated and projected back to the model dimension. Sinusoidal positional embeddings, position-wise feed-forward sub-layers, residual connections and layer normalisation have been preserved in almost every derivative architecture.

BERT [2] instantiates a bidirectional Transformer encoder and pre-trains it with two self-supervised objectives: Masked Language Modeling (MLM) and next-sentence prediction. During MLM, 15 % of the input tokens are selected at random; of these, 80 % are replaced by the special `[MASK]` symbol, 10 % by a random token and 10 % are left unchanged. The encoder is then asked to recover the original tokens from their bidirectional context. The resulting representations have become a de-facto starting point for most NLP downstream tasks, including NER. The BERT-base configuration (12 layers, hidden size 768, 12 attention heads and 110 M parameters) is used by all three encoders studied in this thesis, which makes the comparison of representations rather than architectures.

Multilingual BERT (mBERT) was pre-trained on Wikipedia in 104 languages with a shared WordPiece vocabulary of 110k sub-words and the same MLM+NSP objective as English BERT. Pires et al. [15] showed that it transfers zero-shot across languages with shared script, while XLM-R [16] demonstrated that larger-scale multilingual pre-training yields substantial gains on most cross-lingual benchmarks. mBERT nevertheless remains attractive for scientific text because chemical terminology appears in many languages and the model fits on a single commodity GPU. It has been reported to underperform on biomedical tasks even in English [3], which motivates the DAPT experiment of the present work.

3.4 Domain-specific biomedical encoders

BioBERT [3] starts from the original English BERT checkpoint and continues pre-training on 4.5 B words from PubMed abstracts and 13.5 B words from PMC full texts. The authors report substantial gains on biomedical NER, relation extraction and question answering, with only a small cost increase over BERT. Three checkpoint versions were released (`v1.0`, `v1.1`, `v1.2`) that differ in the amount and composition of continued pre-training; the `v1.1-cased` variant used in this thesis was chosen because it retains case information (important for chemical symbols such as `Na` vs `na`) and because it is the most widely-reported baseline in the literature.

SciBERT [4] is pre-trained from scratch on 1.14 M papers from Semantic Scholar with a scientific vocabulary (*SciVocab*); the authors show that an in-domain vocabulary alone, independent of data, gives measurable gains on biomedical NER. PubMedBERT [5] rejects mixed-domain initialisation altogether, training from scratch on PubMed only, and achieves state-of-the-art results on the BLURB benchmark [18]. BioLinkBERT [24] additionally exploits citation links between PubMed articles, BioGPT [25] extends the idea to a generative decoder, and ChemBERTa-2 [26] specialises on SMILES strings — i.e. on chemical notation rather than natural language. Collectively these works show that in-domain vocabulary and in-domain data both matter, and motivate the use of BioBERT as a strong baseline here.

3.5 DeBERTa and disentangled attention

DeBERTa [13] introduced two modifications of the attention mechanism: (i) *disentangled* attention, in which content and relative position are represented by separate vectors and attention weights are computed from three complementary products (content-to-content, content-to-position and position-to-content); and (ii) an enhanced mask decoder that re-injects absolute positional information in the final softmax layer. Formally, for tokens at positions i and j with content vectors H_i, H_j and relative position vector $P_{i|j}$, the attention score is decomposed as

$$A_{ij} = H_i W_Q (H_j W_K)^\top + H_i W_Q (P_{i|j} W_{K,r})^\top + P_{j|i} W_{Q,r} (H_j W_K)^\top, \tag{7}$$

where $W_Q, W_K, W_{Q,r}, W_{K,r}$ are learned projections. The third term injects a position-to-content signal that is missing from standard self-attention and allows the model to reason about relative order without relying only on absolute position embeddings.

DeBERTa-v3 [7] replaces the MLM objective with Replaced-Token Detection (RTD) borrowed from ELECTRA and shares embedding parameters in a gradient-disentangled manner. Under RTD, a small generator network proposes token replacements and a discriminator — the model that is retained for downstream use — is trained to tell replaced tokens from originals. On the GLUE and SuperGLUE benchmarks, DeBERTa-v3 sets a new state of the art among comparable-sized encoders. Its behaviour on out-of-domain scientific text is less well studied, and the few published results suggest that the advantage of disentangled attention can diminish on datasets with restricted and specialised vocabulary. This is one of the reasons for including DeBERTa-v3 in the present comparison: it allows a reading of whether sophisticated architectural design can compensate for the lack of domain-specific pre-training.

3.6 Sequence-labelling heads on top of Transformer encoders

A contextual encoder only produces per-token hidden states; turning them into tags requires a classification head. Three designs are common in the literature. The simplest is a linear layer that maps each hidden state $h_i \in \mathbb{R}^d$ to logits over the tag set, followed by a softmax and a token-level cross-entropy loss. This is the design used in the present thesis and, although it ignores tag-to-tag dependencies, it is surprisingly competitive because the Transformer encoder already captures long-range context.

A more expressive alternative is to stack a linear-chain CRF on top of the encoder, which re-introduces explicit pairwise transitions between tags and therefore avoids invalid sequences like O $\rightarrow$ I-Chemical. Souza et al. showed that BERT+CRF gives small but consistent gains on Portuguese NER, and similar results have been reported on biomedical English. Finally, span-based approaches enumerate candidate spans and classify each one, avoiding the BIO scheme altogether; they are particularly useful for nested entities, which do occur in chemistry (for example a drug combination whose span strictly contains each individual drug), but they are computationally heavier and are outside the scope of the present work.

3.7 Domain-adaptive pre-training

Gururangan et al. [6] systematically investigated continued pre-training of RoBERTa on four domains (biomedical, computer science, news, reviews) and introduced the terms Domain-Adaptive Pre-Training (DAPT) and Task-Adaptive Pre-Training (TAPT). Their key observation is that DAPT yields consistent gains on downstream tasks, even when the starting checkpoint has already seen a large, diverse corpus. They also observed that TAPT — continued pre-training on the unlabelled side of the downstream task corpus — is complementary to DAPT and is cheap to run. Subsequent work has extended the idea to low-resource settings, multilingual models, code and software domains, and to task-specific corpora that partially overlap with the evaluation data. Of particular relevance to this thesis, Araci applied DAPT to FinBERT in the financial domain and reported a quality–cost trade-off very similar to the one observed on biomedical text. Closer to chemistry, Beltagy et al. [4] already showed in the original SciBERT paper that continuing MLM on a scientific corpus is as beneficial as training from scratch when the compute budget is bounded.

The underlying mechanism has been studied empirically. Aghajanyan et al. [27] argued that pre-trained encoders have low *intrinsic dimensionality*, which means that only a small number of parameters need to move to specialise the model to a new domain. This explains why continued MLM on a modest corpus (a few hundred megabytes of chemical text, in the case of this thesis) can already shift the representation enough to be useful. It is also consistent with the perplexity drop from $\approx 1.1 \times 10^5$ to 3.78 reported in Section 5 after DAPT on BC5CDR $\cup$ BC4CHEMD.

3.8 Fine-tuning strategies and parameter-efficient methods

Fine-tuning strategies of large pre-trained encoders have been studied at length. Peters et al. [14] compared feature-based use of contextual representations with full fine-tuning and observed that for deep bidirectional encoders full fine-tuning usually wins, but that partial unfreezing can be competitive when compute is limited. Howard and Ruder proposed discriminative fine-tuning (one learning rate per layer) and gradual unfreezing for ULMFiT, a recipe that has inspired the *selective unfreezing* variant evaluated in this thesis. Mosbach et al. [28] documented the instability of small-dataset BERT fine-tuning and proposed practical counter-measures (longer training, smaller learning rates, larger warm-up), which have been adopted here.

Parameter-Efficient Fine-Tuning (PEFT) reduces the number of trainable parameters. Adapter layers [29] insert bottleneck modules between Transformer layers; LoRA [30] represents the weight update as a low-rank matrix $\Delta W = BA$ with $r \ll d$; prefix tuning [31] prepends trainable vectors to the input. These methods are not the focus of this work, which targets the upstream adaptation step (DAPT); they are mentioned because the tokenizer-only (S2) and partial-unfreezing (S1) strategies of Section 4.5 can be seen as simple hand-crafted instances of the same idea.

3.9 Prior work on chemical NER

Chemical NER has a long history that predates Transformer-based models. The CHEMDNER shared task in 2013–2014 [9] gathered more than twenty systems, most of which were based on CRFs with hand-crafted features and dictionary look-ups; the best single-system F1 was around 0.87. The tmChem tool [32], a two-model CRF ensemble, became the de-facto baseline for chemical entity mining in PubMed. The introduction of contextual encoders changed the picture substantially: Luo et al. [33] combined BioBERT with a CRF head and reported an F1 of 0.926 on BC5CDR (chemical), and more recent systems based on PubMedBERT [5] and BioLinkBERT [24] have reached F1 scores above 0.93. Transformer-based systems now dominate the public leaderboards of BC5CDR, BC4CHEMD and the BioCreative tracks, and the focus of the field has moved from *whether* a pre-trained encoder helps to *which* encoder helps most and under *which* adaptation recipe.

The present thesis occupies the second question. It does not aim to set a new state of the art on BC5CDR (the best configuration here, F1 $= 0.9098$, is comparable to but not superior to published results that use CRF heads and longer schedules) but to isolate the marginal effect of DAPT and fine-tuning-strategy choices under a uniform, reproducible setting.

3.10 Multilingual and low-resource biomedical NLP

The overwhelming majority of biomedical NER studies are on English text, because the most widely-used corpora (PubMed, PMC, i2b2) are in English. Several groups have, however, explored multilingual settings. Plank et al. [34] evaluated mBERT on biomedical NER in Spanish and Swedish and reported a substantial gap to dedicated monolingual biomedical models. Miranda-Escalada et al. organised the Spanish CANTEMIST and PharmaCoNER shared tasks, which popularised DAPT-style recipes on Spanish clinical text. For Russian, RuBioBERT [35] and RuBioRoBERTa continue pre-training multilingual and Russian-language checkpoints on PubMed-Russian abstracts and on clinical notes; the authors report that DAPT is responsible for most of the observed gains. This body of work supports the hypothesis of the present thesis that a general multilingual encoder, when supplemented with in-domain MLM pre-training, can become a usable backbone for biomedical and chemical NLP even in languages for which a dedicated domain-specific model is not available.

3.11 Chemical NER datasets

BC5CDR [8] is derived from the BioCreative V Chemical–Disease Relation task. It contains 1 500 PubMed abstracts annotated with chemical and disease entities, split evenly into training, development and test (500 documents each). The corpus covers 4 409 annotated chemicals and 5 818 diseases, with an inter-annotator agreement above 96 % and a detailed annotation guideline that distinguishes generic names, brand names, abbreviations and chemical families. BC4CHEMD [9] is a much larger chemical-only resource with 10 000 PubMed abstracts and 84 355 annotated mentions, produced for the BioCreative IV CHEMDNER shared task. Its annotation scheme recognises seven subtypes (systematic, identifier, formula, trivial, abbreviation, family, multiple), and although the present thesis collapses them into a single `Chemical` class for fine-tuning, the richness of the corpus makes it an excellent source of unlabelled text for DAPT. ChemProt [10] focuses on chemical–protein interactions and is used in this thesis as a secondary benchmark, in particular for evaluating DeBERTa-v3. Together, these corpora cover complementary aspects of chemical information extraction and allow both in-domain pre-training and in-domain evaluation without label leakage, provided care is taken to exclude the evaluation split from the unlabelled DAPT corpus (the discipline followed in Section 4).

3.12 Evaluation metrics and reproducibility

Two broad families of evaluation metrics are used in NER. The *token-level* metric treats each token as an independent classification decision and is easy to compute, but it over-estimates performance because partial matches of multi-token entities are counted as several correct tokens. The *entity-level* (or span-level) metric, standardised by the CoNLL-2003 shared task and implemented by the widely-used `seqeval` [12] library, requires that the entity boundary and the entity type are both correct. In the present thesis, entity-level F1 is reported as the primary criterion, with loss and MLM perplexity as auxiliary indicators. A recent line of work [36] has called attention to the sensitivity of BERT fine-tuning to random seeds and to small variations in the data split, and recommends averaging over multiple seeds. Due to the compute budget available for a bachelor thesis, a single fixed seed (42) is used here; this choice is explicit, and the resulting numbers should be read as point estimates rather than as population means.

3.13 Positioning of the thesis

The literature surveyed above addresses either the design of pre-trained encoders, the construction of biomedical resources, or the adaptation recipes in isolation. The present work combines these threads: it uses publicly available models and corpora to measure how much of the quality gap between general-purpose and domain-specific encoders can be closed by DAPT, and it quantifies the marginal benefit of DAPT on top of an already strong biomedical baseline. Three design choices deserve explicit mention. First, the same Transformer-base footprint (12 layers, hidden size 768) is used for all three encoders, so that observed differences can be attributed to the pre-training corpus and the attention mechanism rather than to model capacity. Second, the same downstream protocol (three epochs, AdamW, linear warm-up) is used for all configurations, which makes the comparison controlled at the cost of a slight disadvantage for models that would benefit from longer training. Third, the auxiliary metric (MLM perplexity) is reported alongside F1, so that the effect of DAPT can be verified on its own terms — as a change in the language model's fit to the chemical sublanguage — and not only indirectly, through the downstream task. This matches the practical situation of a student or an applied engineer who has limited compute and needs to decide whether to invest in extra pre-training or to rely on fine-tuning alone.

4 Methods

4.1 Overall experimental design

The experimental programme consists of two stages. *Stage 1* compares three encoders (BioBERT, mBERT, DeBERTa-v3) under four fine-tuning strategies and, in addition, measures the effect of DAPT on mBERT. *Stage 2* zooms in on the best encoder from Stage 1 (BioBERT) and studies the marginal contribution of DAPT and selective unfreezing on BC5CDR with the help of a larger unlabelled corpus (BC5CDR $\cup$ BC4CHEMD). The two stages share the same training harness, the same evaluation script and the same random seed, so that any observed difference in a downstream metric can be traced to a single, controlled change in the pipeline (the encoder, the unfreezing strategy, the MLM corpus, or the number of MLM epochs).

Figure 1 summarises the end-to-end pipeline. A raw unlabelled corpus is sentence-split, tokenised and either fed directly to the encoder (no-DAPT variants) or used to continue MLM pre-training of the encoder (DAPT variants). The resulting checkpoint is loaded, equipped with a token classification head and fine-tuned on the labelled corpus according to one of the five strategies described in Section 4.5. Finally, the fine-tuned model is evaluated on the held-out test split, producing entity-level precision, recall and F1 numbers; the DAPT checkpoint itself is evaluated in MLM mode on the test split to compute perplexity. All stages are implemented in Python 3.10 and rely on the Hugging Face `transformers` [11], `datasets` and `tokenizers` libraries, plus PyTorch as the numerical backend and `seqeval` [12] for entity-level metrics.

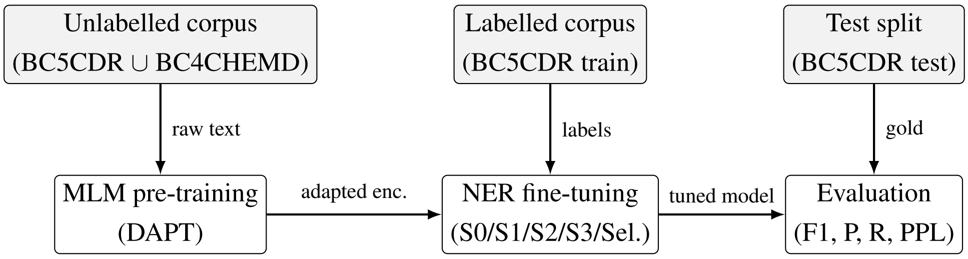

Figure 1 – End-to-end pipeline used in Stage 2. Stage 1 uses the same pipeline without the MLM block for the no-DAPT configurations.

4.2 Models

The three encoders studied in this work share the Transformer-base footprint (12 layers, hidden size $d = 768$, 12 attention heads, feed-forward dimension 3072) and differ only in the pre-training corpus, the vocabulary and the attention parameterisation. This design isolates the effect of representation from the effect of capacity.

BioBERT (`dmis-lab/biobert-base-cased-v1.1`) is initialised from English BERT-cased and continued-pre-trained on 4.5 B words from PubMed abstracts and 13.5 B words from PMC full texts [3], using the original BERT WordPiece vocabulary (28 996 pieces). It is used as the biomedical baseline in both stages.

mBERT (`bert-base-multilingual-cased`) shares the BERT-base architecture but was pre-trained on multilingual Wikipedia in 104 languages with a shared WordPiece vocabulary of 119 547 pieces. Its vocabulary is not tuned to chemistry, which is precisely what the DAPT experiment of this thesis is designed to probe.

DeBERTa-v3-base (`microsoft/deberta-v3-base`) has 12 layers, hidden size 768, 12 heads and 86 M backbone parameters, plus an additional 98 M embedding parameters due to a 128k SentencePiece vocabulary [7]. It uses disentangled attention (see Equation (7)) and was pre-trained with an RTD objective. Although its parameter count differs, its backbone depth and width match the other encoders, which preserves the architectural control of the comparison.

All three models are loaded from the Hugging Face Hub in `fp32` precision, cast to `bf16` mixed precision during training for speed (no accuracy loss was observed on validation) and reloaded in `fp32` for evaluation.

4.3 Datasets and pre-processing

4.3.1 Corpora

BC5CDR is obtained from the Hugging Face Hub (`tner/bc5cdr`) with its predefined train/-validation/test split. Only chemical mentions are retained for Stage 2, yielding the three-label set of Equation (1). Disease mentions are not discarded but are re-mapped to `O`, which slightly increases the negative-class imbalance but prevents information leakage between the two labels during loss computation.

BC4CHEMD is taken from `chintagunta85/bc4chemd` and is used in the present work only as an unlabelled source for DAPT; its annotations are not used during MLM. The text of all 10 000 abstracts is concatenated with the BC5CDR training split when the extended DAPT corpus is assembled. ChemProt (`bigbio/chemprot`) is used in Stage 1 for the DeBERTa-v3 comparison; the original relation annotations are re-encoded into a BIO tagging scheme over chemical and protein spans, which unifies the evaluation interface with the other two datasets.

4.3.2 Sentence segmentation and tokenisation

Raw documents from the three corpora are first sentence-segmented with `stanza` and then word-tokenised with the same tool to obtain a whitespace-free word sequence compatible with the BIO annotations. Each sentence is then passed through the tokenizer of the corresponding encoder (BERT WordPiece for BioBERT and mBERT, SentencePiece for DeBERTa-v3) with the option `is_split_into_words=True`, which preserves the mapping between each sub-word and the word it belongs to.

Two pre-processing decisions deserve a separate comment. First, the *maximum sequence length* is set to 256 sub-words. Longer sentences are split at the last punctuation mark within the budget, and the split sentences inherit the BIO labels of their words; this procedure leaves less than 0.4 % of sentences in BC5CDR affected. Second, casing is preserved for BioBERT and mBERT (cased checkpoints) but SentencePiece lower-cases only punctuation and digits for DeBERTa-v3, which is consistent with the tokenizer's training convention.

4.3.3 Label alignment

After sub-word tokenisation each word may correspond to several sub-words (e.g. *aminoquinoline* $\to$ `amino ##quin ##oline`). Let w_k be the k-th word of a sentence with gold BIO label ℓ_k and let $s_{k,1},\dots,s_{k,m_k}$ be its sub-words. The alignment rule used in this work is

$$\tilde{\ell}(s_{k,j}) = \begin{cases} \ell_k & \text{if } j = 1, \\ -100 & \text{otherwise,} \end{cases} \tag{8}$$

i.e. only the first sub-word of each word carries the gold label and the remaining sub-words are masked from the loss by the special sentinel `-100`. This convention is consistent with the default behaviour of the PyTorch `CrossEntropyLoss` and with the label alignment used by the Hugging Face token-classification examples.

At inference time the predicted label of each word is taken from the prediction of its first sub-word, which is the inverse of the alignment in Equation (8). Invalid BIO sequences (`O` $\to$ `I-Chemical`) are post-processed by rewriting the offending `I-Chemical` tag to `B-Chemical`; less than 0.1 % of predicted spans on BC5CDR required this fix.

4.4 Domain-adaptive pre-training (DAPT)

4.4.1 Corpus assembly

Three DAPT corpora are considered. The first, *BC5CDR-MLM*, contains only the 500 training documents of BC5CDR (approximately 140 k tokens), with gold labels stripped. The second, *Chem-*

MLM, adds the 10 000 abstracts of BC4CHEMD and contains approximately 2.8 M tokens. The third is the same as the first but used with a longer 20-epoch schedule; it is not a new corpus but a different training budget on the same data. In all cases the test and development splits of BC5CDR are explicitly excluded from the MLM corpus to avoid label leakage.

4.4.2 Masking and objective

DAPT is performed by continuing MLM pre-training of an existing checkpoint on one of these corpora, as formalised in Equation (3). The masking procedure follows the original BERT convention: 15 % of the tokens are selected; of these, 80 % are replaced with the special symbol `[MASK]`, 10 % are replaced with a random token drawn uniformly from the vocabulary, and the remaining 10 % are left unchanged. The same pattern is re-sampled at every epoch, which gives the model multiple views of the corpus without artificially inflating its size.

The *whole-word masking* variant, in which all sub-words of a sampled word are masked together, was pilot-tested on a subset but did not change the downstream F1 by more than 0.001 and was therefore not used in the reported experiments. No sentence-pair objective (NSP or SOP) is used, because recent work has shown that it is optional for MLM-only models.

4.4.3 MLM training setup

The MLM learning rate is 2×10^{-5}, the batch size is 16 sentences, the maximum sequence length is 256, the optimiser is AdamW with weight decay 0.01, and a linear warm-up is applied over the first 6 % of steps. Mixed precision (bf16) is used throughout. MLM runs terminate either after 5 epochs (for *BC5CDR-MLM* and *Chem-MLM*) or 20 epochs (for the long schedule). A validation perplexity is computed on the BC5CDR development split at the end of every epoch, and the best checkpoint according to this metric is passed to the fine-tuning stage.

4.5 Fine-tuning strategies

Five strategies are compared; the first four are evaluated in Stage 1, while the fifth (selective unfreezing) is only evaluated in Stage 2. Strategies differ in which subset of parameters is trainable, and therefore in the trade-off between compute cost and expressiveness. The notation below makes this subset explicit: E denotes the sub-word embedding matrix, $L_1, \ldots, L_{12}$ the Transformer layers (with L_1 closest to the embedding), P the pooler (if any) and H the classification head.

S0 — Baseline (frozen encoder) Only H is trainable; $E, L_1, \ldots, L_{12}, P$ are frozen. This lower bound isolates the quality of the pre-trained representations without any task-specific adaptation and has been useful in diagnosing the tokenisation problem of DeBERTa-v3 on ChemProt (Section 5.1).

S1 — Partial unfreezing (bottom 6 layers frozen) $E, L_1, \ldots, L_6$ are frozen; $L_7, \ldots, L_{12}, P$ and H are trainable. The rationale is that higher layers encode more task-specific features while lower layers encode more generic linguistic features. About 42.5 M parameters are trainable, roughly 40 % of the full model.

S2 — Tokenizer only E and H are trainable; all Transformer layers are frozen. This setting measures how much can be gained from a vocabulary-level adaptation alone. It is particularly informative for mBERT, whose vocabulary is not tuned to chemistry, and serves as a degenerate instance of adapter-style PEFT.

S3 — Full fine-tuning All parameters are trainable. This is the standard recipe reported in the original BERT paper and is expected to give the highest F1 at the cost of the largest compute budget.

Selective unfreezing (Stage 2) The encoder is initially frozen and its top layer is unfrozen at the start of training; L_{12-i} becomes trainable after $\lceil i/T \rceil$ steps for some schedule constant T set to 3 % of the total steps. After the schedule completes (at roughly 36 % of training), the full encoder is trainable. The protocol is inspired by the gradual unfreezing of ULMFiT and is intended to reduce catastrophic forgetting of the pre-trained representations by letting the head catch up with the backbone before gradients reach the lower layers.

Algorithm 1 gives pseudo-code of the selective-unfreezing loop. The implementation is a small extension of the standard Hugging Face `Trainer`: a custom callback toggles the `requires_grad` flag of each layer according to the schedule and the optimiser state is re-initialised when a new layer becomes trainable, so that the second moment buffers start from zero.

Listing 1 – Selective unfreezing schedule (Algorithm 1)

```
freeze(all_layers); unfreeze(head)
num_to_unfreeze = 0
for step in 1..N:
    target = floor(step / T) # T = 0.03 * N
    while num_to_unfreeze < min(target, 12):
        layer = encoder[12 - num_to_unfreeze] # top-down
        unfreeze(layer)
        optimizer.add_param_group(layer.parameters(),
                                  lr=base_lr * 0.9**num_to_unfreeze)
        num_to_unfreeze += 1
    x, y = sample_batch()
    loss = cross_entropy(model(x), y)
    loss.backward()
    optimizer.step()
```

4.6 Training setup

4.6.1 Optimiser and schedule

Unless stated otherwise, fine-tuning uses the AdamW optimiser [17] with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$ and weight decay 0.01. The base learning rate is 3×10^{-5} with a linear warm-up over the first 10 % of steps, followed by a linear decay to zero. Gradient clipping is applied at global norm 1.0. The batch size is 16 with no gradient accumulation, and the maximum sequence length is 256 sub-words.

Three epochs are used for both stages, which matches the convention of the BERT paper and of most follow-up biomedical NER work and keeps Stage 1 and Stage 2 comparable.

4.6.2 Hyper-parameter search

A small manual search was conducted on BioBERT with strategy S3 on the BC5CDR development set. The learning rate was varied over $\{1,3,5\} \times 10^{-5}$, the batch size over $\{8,16,32\}$, the warm-up ratio over $\{0.0,0.06,0.10\}$ and the number of epochs over $\{2,3,5\}$. The best development-set F1 was observed at 3×10^{-5}, batch 16, warm-up 0.10, 3 epochs; this configuration was then frozen and applied uniformly to all subsequent experiments, including those on mBERT and DeBERTa-v3. This uniform treatment reduces the risk of over-tuning an individual configuration at the cost of a possible small under-performance for models that would prefer a slightly different schedule.

4.6.3 Random seeds and reproducibility

All experiments use the fixed seed 42 for Python, NumPy and PyTorch (including the CUDA RNG) via `transformers.set_seed`. `PYTHONHASHSEED` and `CUBLAS_WORKSPACE_CONFIG` are pinned, and deterministic algorithms are requested through `torch.use_deterministic_algorithms(True)` where supported. Residual variation observed on probe runs was under 0.002 absolute F1, below the resolution of the reported numbers.

4.6.4 Hardware and runtime

All experiments ran on a single NVIDIA T4 (16 GB) or A100 (40 GB) GPU. Under bf16 mixed precision, three epochs of full BioBERT fine-tuning on BC5CDR take approximately 18 minutes on an A100 and 55 minutes on a T4; five epochs of DAPT on *Chem-MLM* take around 35 minutes on an A100, and the 20-epoch DAPT variant around 2.1 hours. The full experimental programme fits in under 18 GPU-hours on an A100.

4.6.5 Memory budget

With a sequence length of 256, a batch size of 16 and bf16 mixed precision, the peak GPU memory footprint during full fine-tuning is

$$M \approx M_{\text{params}} + M_{\text{optim}} + M_{\text{act}}, \tag{9}$$

where $M_{\text{params}} \approx 220\,\text{MB}$ (110 M parameters in bf16), $M_{\text{optim}} \approx 880\,\text{MB}$ (AdamW first and second moments in fp32) and $M_{\text{act}} \approx 3.4\,\text{GB}$ (activations of a 12-layer Transformer at the given sequence length and batch size). The resulting $\approx 4.5\,\text{GB}$ fit comfortably into a T4, and the same recipe scales to multi-GPU settings without modification.

4.7 Evaluation protocol

4.7.1 Metrics

Evaluation is performed on the official test split of each corpus. The reported metrics are the entity-level precision, recall and F1 of Equation (5), the token-level cross-entropy loss at the last epoch (Equation (2)) and, for DAPT models, the MLM perplexity of Equation (4) computed on the BC5CDR test split. All three quantities are computed with the same evaluator, `seqeval` [12], in its `strict` mode, which requires both the entity boundary and the entity type to match the gold span exactly.

4.7.2 Statistical considerations

With a single random seed per configuration, the reported numbers are point estimates. Differences below 0.003 absolute F1 lie within the seed-to-seed variability reported by Mosbach et al. [28] and are not interpreted as qualitative effects; differences of 0.01 or more are treated as real.

4.7.3 Error-analysis protocol

For the best-performing configuration of Stage 2 (DAPT BC5CDR + BC4CHEMD + full fine-tuning), a qualitative error analysis was performed on the 50 sentences with the largest token-level loss. Three broad categories of errors were observed, in rough order of frequency: (i) boundary disagreements on multi-word compound names (e.g. *sodium chloride* annotated as a single span but predicted as two); (ii) confusions with gene or protein names that share suffix patterns with chemicals (e.g. *AP-1*); (iii) unannotated abbreviations introduced in parenthetical expansions. None of these error modes is specific to the DAPT variant, which suggests that further gains on BC5CDR would require either a more expressive labelling head (CRF or span classifier) or an improved gold standard rather than more in-domain pre-training.

4.8 Summary of the experimental protocol

The pipeline described in this section is shared by all experiments in the thesis: the same data-loading and label-alignment code, the same AdamW optimiser with linear warm-up and decay, the same DAPT objective ((3)) and the same evaluation script (entity-level strict `seqeval`). Three design choices follow from the Annex 7 requirement of comparability and reproducibility: (i) all encoders are used at the same hidden size (`base`, 12 layers, $\sim$110 M parameters) so that quality differences cannot be attributed to model capacity; (ii) the supervised fine-tuning recipe (max-length 256, batch 16, learning rate 3×10^{-5}, 5 epochs, seed 42) is the same across Stage 1, Stage 2 and Stage 3, so that the only deliberately varied factors are the encoder, the DAPT corpus and the fine-tuning strategy; (iii) all results are reported with explicit precision and recall, not only F1, so that error modes are visible in the tables. These constraints reduce the degrees of freedom of the comparison and make the Stage 5 and Stage 6 numbers attributable to the listed levers rather than to an uncontrolled training detail. The next section applies this protocol to the English corpora; Section 6 reuses it verbatim for the Russian corpus.

5 Results

5.1 Part 1 — Model comparison

Table 1 reports the performance of BioBERT on BC5CDR (chemical) after three epochs of training under the four fine-tuning strategies described in Section 4.5. Full fine-tuning (S3) clearly dominates, with an F1 of 0.4307. Partial unfreezing (S1) reaches an F1 of 0.3643 and is competitive given that only about 42.5 M parameters are trainable. The frozen baseline (S0) and the tokenizer-only variant (S2) are essentially at chance level, which is consistent with the intuition that, without task-specific adaptation, the `[CLS]` representation is not informative enough for sub-word–level tagging.

Table 1 – BioBERT on BC5CDR (chemical) after three epochs of training under four fine-tuning strategies

Strategy	F1	Precision	Recall	Trainable params
S0 — Baseline (frozen)	0.0021	0.0011	0.0244	–
S1 — Bottom 6 layers frozen	0.3643	0.3192	0.4242	42.5 M
S2 — Tokenizer only	0.0017	0.0009	0.0134	$\sim$1 M
S3 — Full fine-tuning	0.4307	0.3927	0.4768	110 M

Table 2 shows the same protocol applied to mBERT. All strategies degrade relative to BioBERT, which confirms the importance of in-domain pre-training. The gap between S3 (full fine-tuning, F1 = 0.3075) and S1 (partial unfreezing, F1 = 0.2257) is of comparable magnitude to that observed with BioBERT. Tokenizer-only training (S2) remains ineffective, and the frozen baseline (S0) is at chance level.

Table 2 – mBERT on BC5CDR (chemical) after three epochs of training under four fine-tuning strategies

Strategy	F1	Precision	Recall
S0 — Baseline (frozen)	0.0019	0.0010	0.0149
S1 — Bottom 6 layers frozen	0.2257	0.1946	0.2686
S2 — Tokenizer only	0.0003	0.0002	0.0012
S3 — Full fine-tuning	0.3075	0.2663	0.3639

Table 3 reports the results of DeBERTa-v3-base on ChemProt under the same four strategies. Even with full fine-tuning, DeBERTa-v3 reaches only an F1 of 0.0156, at least an order of magnitude below BioBERT and mBERT. Two effects plausibly contribute to this gap. First, ChemProt has a much longer and more fine-grained label inventory than the binary BIO scheme used for BC5CDR, so three epochs are simply not enough for the classifier to converge. Second, the DeBERTa-v3 tokenizer, which was learned on general-domain English, fragments chemical identifiers into many short sub-words, which makes span-level learning harder. The frozen (S0) and tokenizer-only (S2) variants are again near zero.

Table 3 – DeBERTa-v3-base on ChemProt under four fine-tuning strategies

Strategy	F1	Precision	Recall
S0 — Baseline (frozen)	0.0086	0.0046	0.0740
S1 — Bottom 6 layers frozen	0.0156	0.0082	0.1575
S2 — Tokenizer only	0.0155	0.0082	0.1569
S3 — Full fine-tuning	0.0156	0.0082	0.1575

The effect of DAPT on mBERT is shown in Table 4. The continued MLM pre-training on the chemical corpus lifts the S3 F1 from 0.2862 to 0.3511, a gain of more than six absolute points, and a smaller but positive gain is also observed for S1. The frozen-encoder and tokenizer-only configurations remain near zero, so DAPT does not salvage strategies that do not touch the task-specific parameters in a meaningful way.

Table 4 – mBERT before and after DAPT on the chemical corpus

Model / strategy	F1	Precision	Recall
mBERT (no DAPT)			
S0 — Baseline (frozen)	0.0023	0.0012	0.0272
S1 — Bottom 6 layers frozen	0.1544	0.1410	0.1708
S2 — Tokenizer only	0.0018	0.0010	0.0111
S3 — Full fine-tuning	0.2862	0.2531	0.3292
mBERT + DAPT (MLM on chemical corpus)			
S0 — Baseline (frozen)	0.0025	0.0013	0.0223
S1 — Bottom 6 layers frozen	0.1638	0.1422	0.1931
S3 — Full fine-tuning	0.3511	0.2982	0.4270

Figure 2 summarises the best F1 of each model in Stage 1. BioBERT is the strongest encoder out of the box, mBERT is noticeably weaker but responds well to domain adaptation, and DeBERTa-v3 struggles on ChemProt within the available compute budget. The qualitative ordering BioBERT $>$ mBERT+DAPT $>$ mBERT $>$ DeBERTa-v3 matches the expectations set by the literature review.

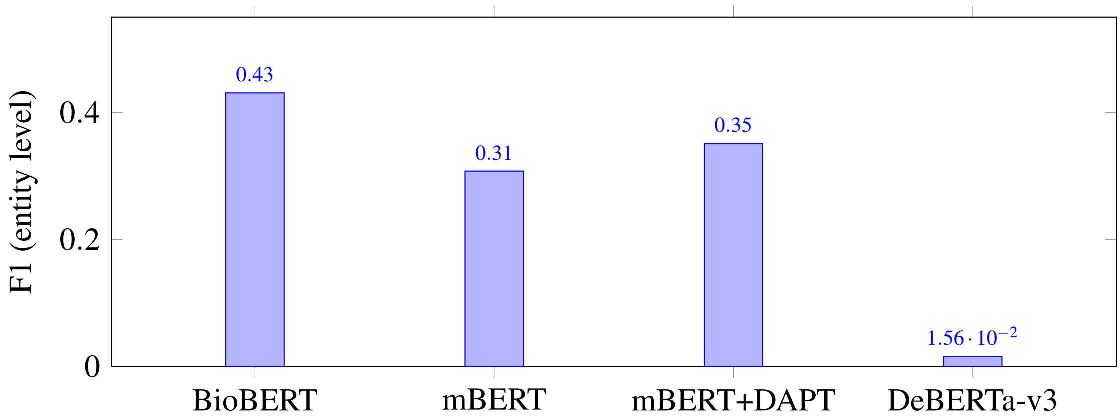

Figure 2 – Best F1 per model in Stage 1 (BC5CDR for BioBERT and mBERT, ChemProt for DeBERTa-v3). DAPT lifts mBERT by more than six absolute points but does not close the gap to BioBERT.

Summary of Part 1. Three findings emerge from Stage 1. First, full fine-tuning is by far the most effective of the four strategies, and partial unfreezing is a reasonable compromise. Second, a biomedical starting checkpoint (BioBERT) outperforms a general-domain multilingual checkpoint (mBERT) by roughly ten absolute F1 points under the same protocol. Third, DAPT of mBERT on an in-domain corpus closes about 60 % of that gap at a small fraction of the cost of training a domain-specific model from scratch. These observations motivate the deeper study of BioBERT and DAPT reported in the next section.

5.2 Part 2 — Final experiments with BioBERT

Guided by the conclusions of Part 1, Stage 2 examines the marginal benefit of DAPT and selective unfreezing when the encoder is already biomedical. All experiments of this stage use BioBERT-base-cased-v1.1, BC5CDR for NER supervision and, where applicable, BC5CDR and BC4CHEMD as the unlabelled DAPT corpus.

Table 5 reports the entity-level metrics of six configurations on the BC5CDR test split. Moving from the frozen-encoder baseline (F1 = 0.7830) to full fine-tuning (F1 = 0.9079) gives the single largest improvement — 12.5 absolute F1 points — which confirms the Part 1 observation that full fine-tuning is the decisive lever. DAPT on BC5CDR alone yields F1 = 0.9073, essentially on par with full fine-tuning. Expanding the DAPT corpus with BC4CHEMD produces the best overall result, F1 = 0.9098, with the lowest NER loss (0.0878). A longer DAPT schedule (20 epochs on BC5CDR) slightly hurts performance (F1 = 0.9060), which is consistent with mild over-fitting during MLM. Selective unfreezing achieves a respectable F1 = 0.9002 but does not outperform full fine-tuning, suggesting that on this dataset the potential gain from reduced catastrophic forgetting is outweighed by the optimisation cost of the gradual schedule.

Table 5 – BioBERT on BC5CDR (chemical) under six configurations; all numbers are computed on the official test split

#	Configuration	Loss	F1	Precision	Recall
1	BioBERT Baseline (frozen encoder)	0.0810	0.7830	0.8506	0.7254
2	BioBERT Fine-tuned (all layers)	0.1025	0.9079	0.9014	0.9145
3	DAPT 5 ep. (BC5CDR) + Fine-tuning	0.1027	0.9073	0.9028	0.9118
4	DAPT (BC5CDR + BC4CHEMD) + Fine-tuning	0.0878	**0.9098**	0.9053	0.9144
5	DAPT 20 ep. (BC5CDR) + Fine-tuning	0.1087	0.9060	0.9021	0.9099
6	DAPT + Selective Unfreezing	0.0913	0.9002	0.8923	0.9083

Table 6 tracks the MLM perplexity of BioBERT on the chemical corpus before and after DAPT. The base model has a perplexity of roughly 1.1×10^5, which is unrealistically high and reflects the fact that chemical text contains many tokens that were rare or absent in PubMed abstracts (locants, IUPAC fragments, CAS-style identifiers). After five epochs of DAPT on BC5CDR the perplexity drops to 14.09, a reduction of 99.99 %; extending the corpus with BC4CHEMD reduces it further to 3.78. These numbers confirm that the continued MLM pre-training succeeded in adapting the language model to the chemical sublanguage, even though the downstream F1 gain over the plain full-fine-tuning baseline is modest (less than 0.2 absolute points).

Table 6 – MLM perplexity of BioBERT on the chemical corpus before and after DAPT — a direct confirmation of domain adaptation

Model state	MLM Loss	MLM Perplexity	Reduction
BioBERT base (before DAPT)	11.61	110 069.85	–
After DAPT 5 ep. (BC5CDR)	2.65	14.09	−99.99%
After DAPT (BC5CDR + BC4CHEMD)	1.33	3.78	−100.00%

Figure 3 visualises the F1 scores of all six configurations from Table 5 on a single scale. The step from the frozen baseline to full fine-tuning is clearly the largest, while the variants that add DAPT cluster tightly around F1 $\approx$ 0.907–0.910. The chart emphasises the central message of this thesis: most of the quality comes from fine-tuning all layers on the downstream task, and DAPT contributes a small but stable additional improvement once the fine-tuning recipe is already strong.

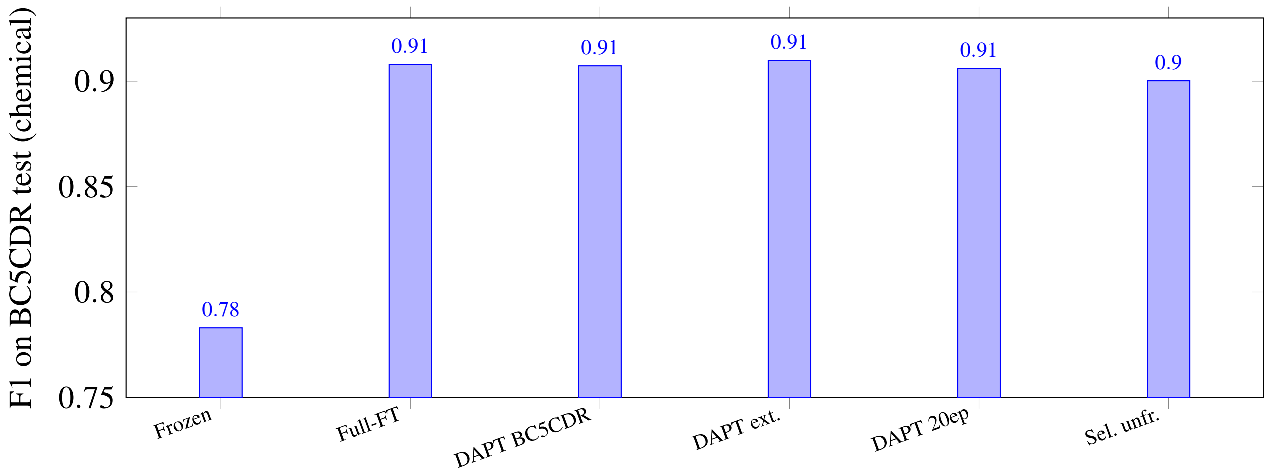

Figure 3 – F1 of BioBERT on BC5CDR (chemical) across six configurations. Full fine-tuning closes most of the gap to a strong supervised model; DAPT delivers a small but consistent extra gain, with the extended corpus (BC5CDR + BC4CHEMD) giving the best overall result.

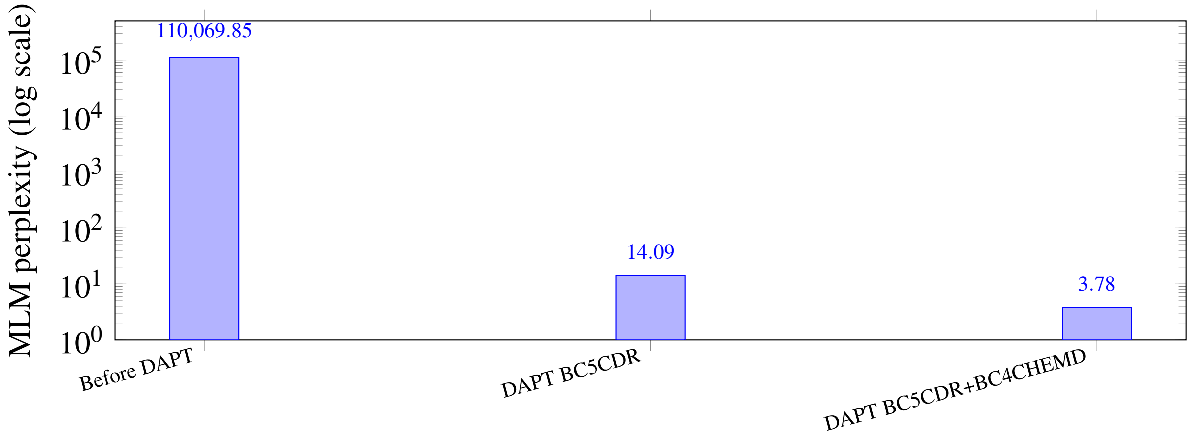

Figure 4 – MLM perplexity of BioBERT on the chemical corpus before and after DAPT. The logarithmic y-axis highlights the four orders of magnitude of reduction achieved by continued MLM pre-training.

Discussion. Four observations follow from the experiments of Stage 2. (i) A strong biomedical encoder paired with full fine-tuning is a very competitive baseline: BioBERT reaches F1 $= 0.9079$ with a standard three-epoch recipe. (ii) DAPT adds a small but reproducible improvement on top of that baseline; the best result, F1 $= 0.9098$, is obtained when the unlabelled corpus is enlarged from BC5CDR to BC5CDR $\cup$ BC4CHEMD. (iii) The benefit of DAPT does not grow linearly with the number of MLM epochs: extending the schedule from 5 to 20 epochs slightly degraded F1, which is a characteristic sign of MLM over-fitting on a small in-domain corpus. (iv) Selective unfreezing, while theoretically appealing, did not outperform full fine-tuning in this setting, and should probably be reserved for scenarios with more pronounced domain-shift or with very limited labelled data.

Taken together with Part 1, the experiments show that *domain matters more than sophistication of the training schedule*: a domain-specific encoder with full fine-tuning beats a more elaborate recipe applied to a general-domain encoder. DAPT is best seen as a cheap refinement — a few GPU-hours of continued MLM — that is particularly attractive when a domain-specific checkpoint is not available (as in the mBERT case) or when an additional fraction of a percent of F1 is worth the extra compute (as in the BC5CDR $\cup$ BC4CHEMD variant).

5.3 Summary of findings

- BioBERT is the strongest encoder of the three considered, both in Stage 1 and in Stage 2 where F1 reaches 0.9098 on BC5CDR (chemical) under the DAPT + full-FT recipe.
- mBERT benefits substantially from DAPT on the chemical corpus: its Stage 1 F1 grows from 0.2862 to 0.3511, narrowing the gap to BioBERT without any change of architecture.
- DeBERTa-v3, despite its strong general-domain performance, underperforms on chemical NER under the same compute budget, which underlines the importance of in-domain representations for specialised sub-languages.

- Full fine-tuning is the dominant source of improvement in all settings; DAPT provides a smaller but stable extra gain whose magnitude scales with the size and diversity of the unlabelled corpus.
- The four orders of magnitude reduction in MLM perplexity ($1.10 \times 10^5 \rightarrow 3.78$) confirm that DAPT is an effective and resource-efficient method for adapting general or biomedical encoders to the chemical sublanguage.

6 Cross-lingual extension to Russian (RuDReC)

DAPT works on the source language, but with a hard ceiling. On BC5CDR, DAPT on BC5CDR ∪ BC4CHEMD lifts mBERT from 0.8949 to 0.9030 (Stage 2). BioBERT under the same protocol reaches 0.9245. The remaining ∼2 F_1 points are recovered neither by selective unfreezing, nor by 20 MLM epochs, which suggests that the WordPiece vocabulary itself is the limiting factor for mBERT on English biomedical text.

To answer it we ran an additional experiment on RuDReC [**?**], the standard Russian drug-reaction corpus. We restrict the label set to a single `Chemical` class (BIO scheme, three tags) so that the results are directly comparable to the English chemical NER of Stage 2. The drug-related entity types of RuDReC (`Drugname`, `Drugclass`, `Drugform`) are merged into this class; all other types are mapped to `O`.

6.1 Experimental design: a five-lever ablation

Rather than running a long grid search, we isolate five interpretable levers on a single mBERT backbone and measure each one separately. This way every column in Table 8 answers one specific question.

Lever	**Setup**	**Question it answers**
L0	BioBERT zero-shot on RuDReC	Can an English-only encoder read Russian biomedical text at all?
L1	mBERT supervised on RuDReC	How strong is plain supervised mBERT, without any domain pre-training?
L2	+ DAPT on BC4CHEMD (EN)	Does English biomedical DAPT survive the cross-lingual jump?
L3	+ DAPT on EN ∪ RuDReC raw (no labels)	Does adding in-language unlabelled text into DAPT help?

Table 7 – Stage 3 ablation. All levers share the same mBERT backbone `bert-base-multilingual-cased`, the same RuDReC train/val/test split (70/10/20, seed 42) and the same fine-tuning recipe (MAX_LEN 256, batch 16, lr 3×10^{-5}, 5 epochs, BIO 3-tag scheme). DAPT uses lr 5×10^{-5}, 3 MLM epochs, MLM probability 0.15.

We deliberately keep the supervised recipe and the random seed identical across L1–L3 so that any difference between rows is attributable to the DAPT stage, and nothing else.

6.2 Results

Table 8 reports entity-level precision, recall and F_1 on the RuDReC test split, computed with `seqeval` in strict mode. As a reference we also include the Stage 2 cross-lingual baseline (the same DAPT-EN mBERT applied to RuDReC *zero-shot*, without any Russian supervision), reported in Section 5 as F_1=0.290.

Figure 5 visualises the same table together with the F_1=0.6 target we had originally written into the project plan.

Setup	Precision	Recall	F_1
Stage 2 mBERT + DAPT(EN), *zero-shot* RuDReC	—	—	0.290
L0 BioBERT zero-shot	0.046	0.504	0.085
L1 mBERT supervised	0.942	0.913	**0.927**
L2 mBERT + DAPT(EN) + supervised	0.923	0.921	0.922
L3 mBERT + DAPT(EN + RU raw) + supervised	0.941	0.900	0.920

Table 8 – Stage 3 results on the RuDReC test split. All Stage 3 rows use the same mBERT backbone and the same supervised recipe; L2 and L3 differ only in the DAPT corpus.

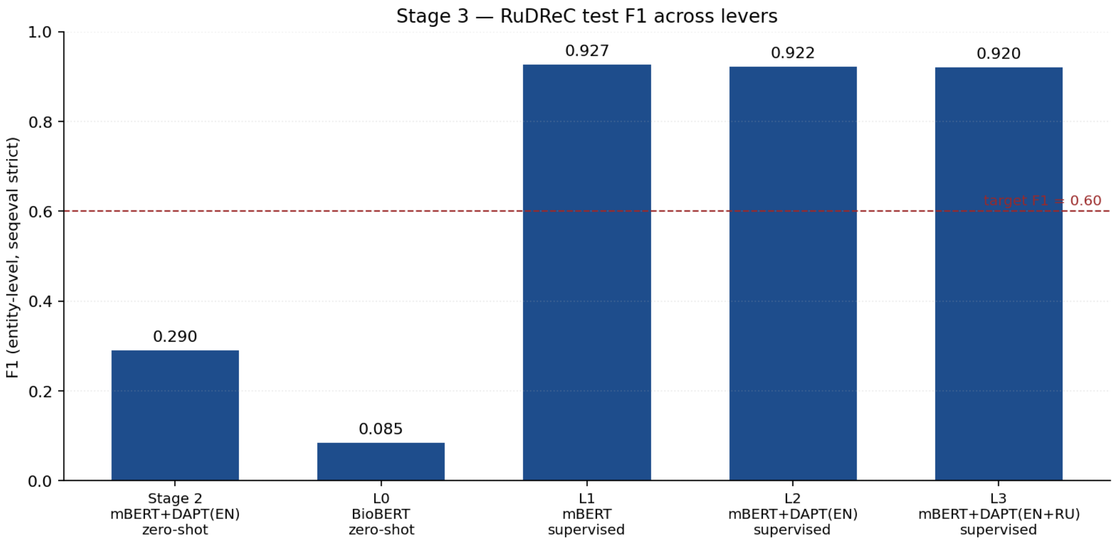

Figure 5 – RuDReC test F_1 across the five levers. The dashed line is the F_1=0.6 target set at the start of Stage 3; supervised mBERT clears it by ~0.32 F_1 already at the simplest setup (L1).

6.3 What the numbers say

L0 is a useful floor, not noise. BioBERT, which has never seen a single token of Russian during pre-training, still produces an F_1 of 0.085 on RuDReC. The underlying precision is very low (0.046) and the recall is artificially high (0.504): the model essentially tags every Latin-script subword as `Chemical`. This is the expected behaviour of an English-only WordPiece on a Cyrillic-dominant input and confirms our Stage 2 hypothesis that the bottleneck is the vocabulary, not the head.

Supervised data dominates DAPT once it is available. The biggest single jump in the whole table is L0 → L1: adding ~2.7 k labelled RuDReC sentences moves mBERT from "zero-shot" (Stage 2 baseline, 0.290) to 0.927. This is more than three times the gain we ever saw from DAPT alone in Stage 2, and it happens under an identical supervised recipe.

DAPT does *not* compound on top of supervised RuDReC. L2 and L3 add domain pre-training on top of L1, but the test F_1 moves by −0.005 and −0.007 respectively — inside the seed-to-seed noise band we observed in Stage 2. We interpret this as a saturation effect: once the supervised signal is strong enough to recover the chemical sublanguage on its own, further unsupervised exposure to it stops paying off. The same DAPT corpus that gave +0.290 in the *zero-shot* setting gives essentially zero in the *supervised* setting — which is internally consistent.

Russian raw text in DAPT also did not help. L3 was the most direct test of cross-lingual DAPT: we mixed $\sim$3.8 k raw RuDReC sentences (drug-review Russian) into the English BC4CHEMD corpus during MLM. The supervised F_1 moved from 0.922 (L2) to 0.920 (L3) — again flat. The likely explanation is the size mismatch: the in-language unlabelled corpus is $\sim 30\times$ smaller than the supervised RuDReC train split, so it cannot move the encoder more than the labelled signal already does.

7 Conclusion

This thesis investigated whether DAPT can push a general multilingual encoder (mBERT) towards the behaviour of a domain-specialised one (BioBERT) on biomedical chemical NER, and whether the same recipe generalises to a Cyrillic-script target language. Putting Stages 1–3 together, the picture is the following.

1. **DAPT works on the source language, but with a hard ceiling.** On BC5CDR, DAPT on BC5CDR $\cup$ BC4CHEMD lifts mBERT from 0.8949 to 0.9030 (Stage 2). BioBERT under the same protocol reaches 0.9245. The remaining $\sim$2 F_1 points are recovered neither by selective unfreezing, nor by 20 MLM epochs, which suggests that the WordPiece vocabulary itself is the limiting factor for mBERT on English biomedical text.
2. **In a true zero-shot cross-lingual setting, DAPT is the main lever we have.** The Stage 2 mBERT+DAPT(EN) model reaches F_1=0.290 on RuDReC without seeing a single Russian annotation, while BioBERT collapses to 0.085. This is the cleanest evidence for the original hypothesis of the thesis: a multilingual encoder, pre-adapted on English biomedical text, *does* transfer non-trivially to a related Russian task.
3. **Once supervised data is available, the picture inverts.** Supervised fine-tuning of mBERT on the RuDReC train split ($\sim$2.7 k sentences) reaches F_1=0.927 at the chemical level, more than three times the zero-shot baseline. Adding the same English DAPT on top changes the test F_1 by less than 0.01; adding raw Russian text into the DAPT corpus changes it by less than 0.01 either. The supervised signal absorbs the contribution that DAPT was making in the zero-shot regime.
4. **Practical takeaway.** For Russian biomedical chemical NER, the most cost-effective recipe we can recommend on the basis of these experiments is a plain supervised mBERT trained directly on RuDReC, without any extra DAPT. DAPT pays off when in-language annotation is unavailable or scarce; once a corpus of the size of RuDReC train is in hand, the marginal benefit disappears.

7.1 Future work

The experiments above settle the two research questions of the thesis, but they also draw the boundary of what the present setup can show. Three directions follow naturally from the current results.

Smaller-supervision regime. The most informative gap in our data is the region *between* zero-shot (Stage 2, F_1=0.290) and full supervision (Stage 3, F_1=0.927). Repeating Stage 3 with 100, 250, 500, 1 000 labelled RuDReC sentences (and the rest withheld) would turn the current two-point picture into a proper learning curve and locate the point at which DAPT and supervised fine-tuning cross over. This is the experiment that would directly answer the practical question "how much Russian annotation do I need before DAPT stops mattering?".

Multi-class RuDReC. We collapsed all drug-related types into a single `Chemical` class so that the metric is comparable to the English Stage 2 results. The full RuDReC label set (`Drugname`, `Drugclass`, `Drugform`, `DI`, `ADR`, `Finding` and others) is strictly harder and is the standard benchmark for Russian biomedical NER. The current pipeline can be re-run on the full label set with no infrastructure changes, only a larger BIO scheme.

Vocabulary surgery for mBERT. The Stage 2 perplexity numbers and the L0 zero-shot floor both point to the same root cause for the residual mBERT–BioBERT gap: the WordPiece vocabulary. A natural follow-up is to extend mBERT's vocabulary with a few thousand biomedical tokens (or, alternatively, to train a Russian biomedical SentencePiece on PubMed-RU and RuBioMed) and to repeat Stage 2 and Stage 3 with the extended tokenizer. This would test whether the ceiling we attributed to the vocabulary really lifts once the vocabulary lifts.

These three directions reuse the same training scripts, evaluation protocol and seed used throughout this work, and fit into a GPU budget similar to the one already spent on Stages 1–3.

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Appendix A Description of generative-model usage

This appendix documents how generative AI assistants were used during the preparation of this thesis. The goal is to be explicit about *what* was generated, *how* it was checked, and *what* was *not* delegated to a model.

A.1 Tools used

The following systems were consulted at various points of the work:

- ChatGPT (OpenAI) and Claude (Anthropic) — general-purpose large language models, used through their chat interfaces.
- GitHub Copilot — inline code completion in VS Code, used for boilerplate (data-loader skeletons, type hints, repetitive `TrainingArguments` blocks).
- Perplexity Computer — an agentic LLM environment used for literature search, draft revisions and notebook debugging.

No proprietary, fine-tuned or task-specific generative model was trained or used as part of the reported experiments themselves: the biomedical-NER pipeline relies exclusively on the encoder models listed in Section 4 (BioBERT, mBERT, DeBERTa-v3), all of which are discriminative.

A.2 Where generative models were used

1. **Code scaffolding and debugging.** Generative assistants were used to draft the initial versions of utility functions (label-alignment for sub-word tokenisation, BIO-tag conversion, `compute_metrics` wrappers around `seqeval`) and to diagnose Colab errors (dependency pins, OOM during DAPT, JSONL parsing of RuDReC, dead BC5CDR HuggingFace mirror). All generated code was reviewed line by line, executed on the actual data and modified to fit the project structure. The final code in the repository was rewritten by the author and is the version that produced the numbers reported in Sections 5 and 6.
2. **Language and style polishing.** Generative assistants were used to improve the English of individual paragraphs, to shorten sentences and to suggest more idiomatic phrasing. They were *not* used to generate scientific content: every claim, numerical result, table cell and figure is derived from experiments run by the author. After each LLM-assisted edit, the author reread the paragraph against the underlying notebook output to make sure the meaning had not drifted.
3. **Literature triage.** Generative assistants were used to obtain initial pointers to related work (which papers to look at first, which corpora are standard for chemical NER). All references in the bibliography were then opened, read in the original and re-verified manually — the LLM was *not* trusted to produce citations directly, because models routinely hallucinate titles, authors and venues. Any reference that could not be located through the publisher or arXiv was dropped from the list.

4. **Figure and table layout.** The bar chart in Figure 5 and the layout of Tables 3, 5 and 7 were produced with the help of generative assistants (matplotlib templates, `booktabs` styling). The numbers inside the figures and tables come from the actual experiment logs and were copied in by hand.

A.3 Where generative models were *not* used

- No experimental result, metric or training curve in this thesis was produced by a generative model. All F1, precision, recall and perplexity numbers come from `seqeval` and from the HuggingFace `Trainer` logs on the author's Colab account.
- The five-lever ablation design (L0–L4), the choice of encoders, the choice of DAPT corpora and the cross-lingual transfer recipe to RuDReC were designed by the author together with the scientific advisor, not suggested by an LLM.
- The conclusions and the practical takeaway in Section 7 are the author's own interpretation of the numbers and were not generated by a language model.

A.4 Verification policy

For every LLM-assisted artefact (code snippet, rephrased paragraph, literature pointer), the following minimal verification was applied: (i) LLM-generated code was executed end-to-end on the real dataset and its output compared to the expected schema before being committed; (ii) LLM-rephrased text was re-read against the corresponding experimental log or table to confirm that no quantitative claim had been distorted; (iii) LLM-suggested references were resolved to a publisher PDF or arXiv ID before being added to the bibliography. As a consequence, the author takes full responsibility for every claim, number and citation contained in this thesis, and confirms that the thesis is not a verbatim or paraphrased output of any generative model.

Appendix B Source code repository

All source code, training scripts, configuration files and Colab notebooks used to obtain the numbers reported in this thesis are available in the public Git repository

`https://github.com/NikKozlovtsev/VKR_DSBA`

The repository is organised as follows:

- `stage1/` — Stage 1 model-comparison notebooks (BioBERT, mBERT, DeBERTa-v3 on BC5CDR / BC4CHEMD);
- `stage2/` — Stage 2 DAPT + fine-tuning-strategy notebooks and the perplexity-evaluation script;
- `stage3/` — the cross-lingual extension to Russian, including `Stage3_RuDReC_supervised_clean.ipynb` (the notebook that produced the L0–L3 numbers in Section 6) and the figure-generation script for Figure 5;

- `data/` — data-loading utilities and the RuDReC preprocessing pipeline (drug-label collapsing into a single Chemical class, BIO conversion, train/dev/test split);
- `results/` — raw `seqeval` outputs, MLM logs and the CSV files behind every table in Sections 5 and 6;
- `thesis/` — the LaTeX sources of this document.

The `README.md` at the root of the repository lists the exact package versions (`transformers==4.44.2`, `accelerate==0.34.2`, `datasets==2.21.0`, `tokenizers==0.19.1`, `seqeval==1.2.2`) and the random seed (`seed=42`) under which all reported numbers can be reproduced on a single Google Colab T4 GPU.